



INSTRUCTIONS for COLUMN REGENERATION

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1 REGENERATION

Purpose of Regeneration: The purification column contains a mixture of copper-based catalyst and molecular sieve to absorb oxygen and moisture, respectively. Over time, these materials become saturated and lose their ability to absorb oxygen and moisture. regeneration restores these materials to their original state, allowing continuous scrubbing of oxygen and moisture.

Recommended Frequency: It is recommended to perform regeneration every three years to maintain optimal performance. If you notice reduced performance if your glovebox it may be necessary to regenerate more often.

Warning for Oxygen Levels: The final step (Cooling) exposes the freshly regenerated purifier material to the PureLab HE atmosphere. Ensure that the PureLab HE atmosphere contains less than 50 ppm oxygen before starting the regeneration. If the oxygen level is above 50 ppm, perform a purge during regeneration.

NOTE: Ensure that you do not expose the purifier materials to atmosphere during the regeneration sequence. If you have any questions or need assistance, contact Inert Support.

1.1 SINGLE-COLUMN PURIFICATION SYSTEM REGENERATION PROCEDURE

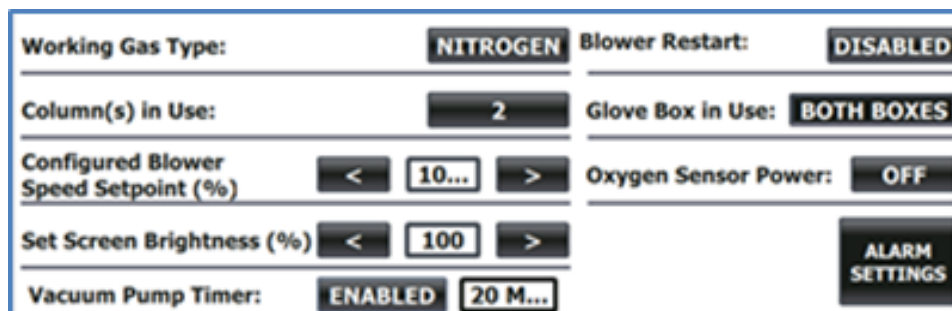
1. **Connect Regeneration Gas Cylinder:** Connect a supply of Nitrogen and 3 – 7% Hydrogen gas mix to the “Regen Gas” port on the rear of the Gas Management module. Set pressure to 7 PSI.
2. **Turn Off Blower:** On the Main Menu screen, turn off the Blower. This automatically closes the purifier column valves.
3. **Initiate Regeneration Sequence:** Tap the "REGEN START" button on the Regen Setup screen
4. **Confirm Column Valve Closure:** Ensure the column valves are closed by observing if the yellow plungers on top are depressed. Once you’ve visually confirmed the valve closure, press the "Yes" button next to "Are the Column Valves Closed?" A "click" sound indicates the solenoid valves opening to allow regeneration gas flow.
5. **Adjust Regeneration Flow:** Adjust the flow control on the regeneration flow meter (located on the front of the Gas Purifier) to 25-30 SCFH.
6. **Verify Flow:** Ensure that gas is flowing out of the regeneration exhaust of the system. Then press "Yes" next to "Is Flow OK?"
7. **Automatic Progression Through Steps:** The system automatically progresses through regeneration steps 1 to 5. At the conclusion of the regeneration sequence, the blower automatically turns back on if enabled in the System Setup screen.

NOTE: Ensure proper safety measures during the regeneration process. The purifier column becomes very hot. Ensure regeneration exhaust gas is properly ventilated.

1.2 DUAL-COLUMN PURIFICATION SYSTEM REGENERATION PROCEDURE

For dual-column purification systems, the regeneration procedure allows one purifier column to regenerate while the other column remains in use. This ensures continuous operation.

1. **Column Selection:** From the System Setup screen, identify the column currently in use in the "Column(s) In Use" field. If you wish to regenerate column 1, select column 2 in the "Column(s) in Use" field.



Working Gas Type:	NITROGEN	Blower Restart:	DISABLED
Column(s) In Use:	2	Glove Box in Use:	BOTH BOXES
Configured Blower Speed Setpoint (%)	< 10... >	Oxygen Sensor Power:	OFF
Set Screen Brightness (%)	< 100 >	ALARM SETTINGS	
Vacuum Pump Timer:	ENABLED 20 M...		

Figure 1 - System Setup Screen

2. **Regeneration Steps:** Follow steps 1, 3, 4, 5, and 6 of the single-column regeneration process above. Ensure the blower remains switched on during the regeneration of a dual-column system.

1.3 SOLVENT COLUMN REGENERATION

NOTE: The Blower does not need to be turned off.

NOTE: You will need a 1/8th" Allen wrench to loosen the screw that secures the sliding panel.

1. Connect high purity inert gas to the regen gas connection on the rear of the Gas Purifier. The Inert gas should be 99.998 nitrogen or argon. Ensure that the connections are tightened correctly. Open the regulator and set it to 7 psi.
2. Open the bypass valve (labeled A in FIGURE 2 below). This is located in the Gas Purifier. The top must be removed to access the valve.
3. Close the isolation valves (labeled B & C) connected to the solvent column. This is located in the Gas Purifier. The top must be removed to access valve.
4. From the Main Menu screen tap the Regen Setup button. On the Regeneration Setup screen tap the Regeneration Type button to select Solvent Regeneration.
5. Tap the Regen Start button.
6. Tap the Yes button next to "Confirm Column Valve is Closed" A "click" will be heard as the solenoid valves open to allow regeneration gas to flow.

- 7 Adjust needle valve on the regeneration flow meter located on the front of the Gas Management module to set the flow to 25-30 scfh on the graduated scale on the flow meter.
- 8 Verify that gas is flowing out of the exhaust line. If it is flowing, tap Yes next to “confirm Flow Status?” At this point the regeneration gas stops flowing. Step 1. Heating begins. The Regen Status shows as Heating and the timer begins to count up. The Regeneration Status is also shown on the Main Screen.
- 9 The system automatically progresses through the regeneration steps 1 to 5.
- 10 Upon completion of the solvent regeneration, open the column isolation valves (B & C), then close the Solvent by-pass valve (A).

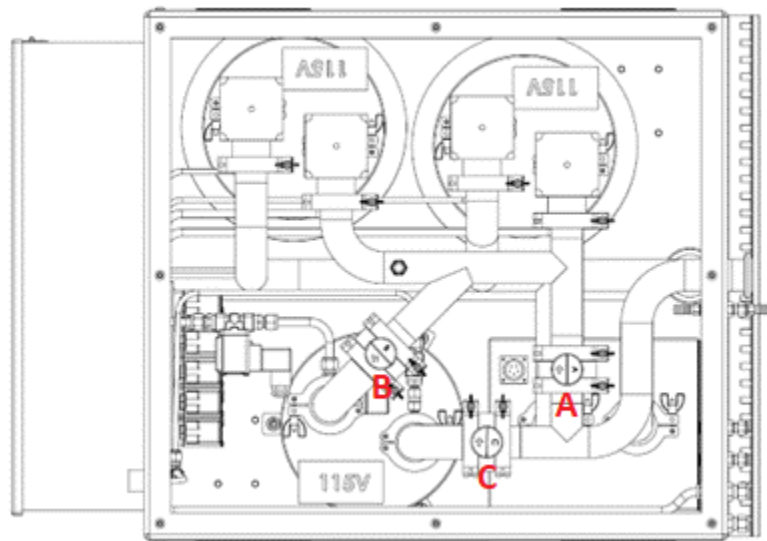


Figure 2 - Top view of Gas Purifier with Regenerable Solvent Removal

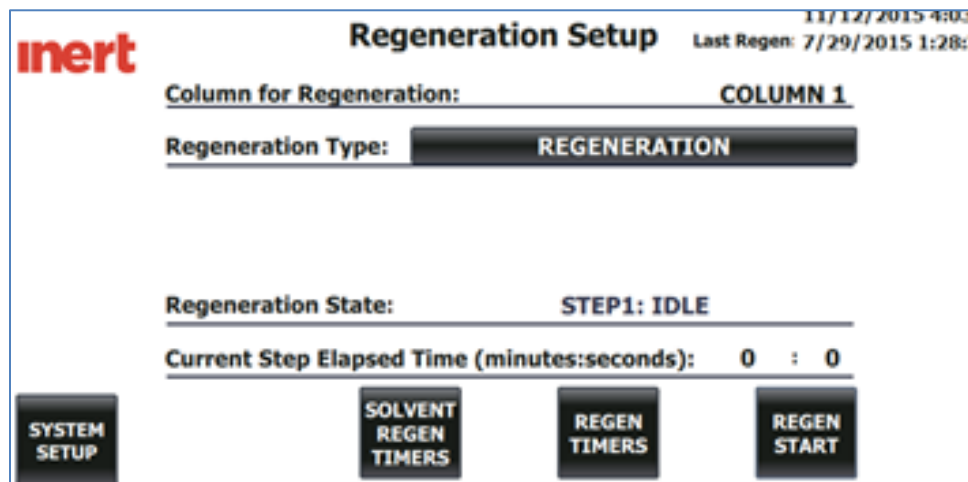


Figure 3 - Regeneration Setup Main

